DataBase Assignment

EE 470 - Introduction to the Internet of Things- Fall / Version: V2

NAME Lucas Loughner DATE: 10/20/26

This project must be done individually!

Link to this document:
https://docs.google.com/document/d/1zhLaeiWjtqfpCTeJzd_n543yl6_SxahBxWd4lHh4srY/edit?usp=sharing

Please make sure you submit your assignment individually!

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Assignment Overview

The purpose of this exercise is to become familiar with MySQL database and its commands.

1. Learning Objectives

- Set Up an online web page
- Learn about SQL
- Practice SQL commands
- Create a Table for sensor data
- Using Github.

2. Required Materials & Reference Information

- Signup for https://hostinger.com
- database tutorial available on the course GitHub
- Tutorial on creating an account with hostinger.com
- You can find the longitude and latitude format here
- Good tutorial: https://www.tutorialspoint.com/mariadb/mariadb_introduction.htm

3. Experiment

This experiment has TWO parts.

Part 1:
In this part of the experiment you need to signup for an account at https://hostinger.com
This will allow you to have your personal web page and database.

Part 2:
In the second part of the experiment you will need to access the database using a PHP file and visualize the data using the Chartjs library.

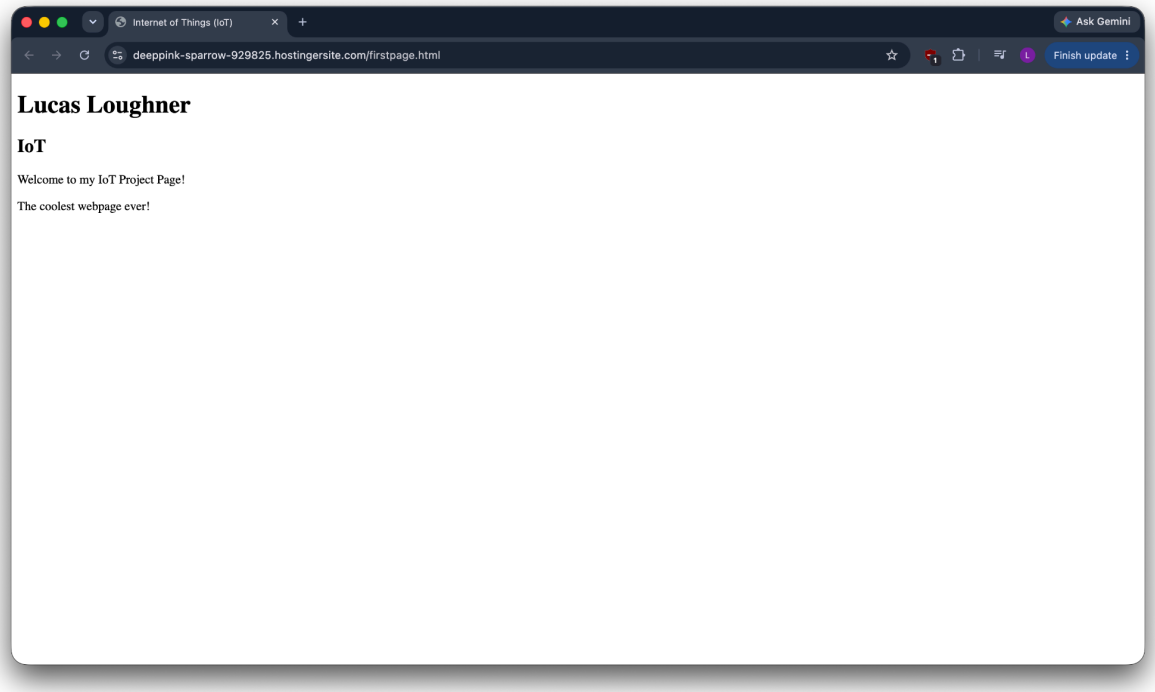
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PART 1

1.1. Create an Account

A- Carefully review the Tutorial on creating an account with hostinger.com. Create a very simple webpage, as described in the tutorial. Make sure your web page is working. Add your name to your webpage. We will call this web page Internet of Things (IoT). We will use this web page to place links to all your future assignments.

Link to your webpage
https://lucasloughner.site/
Snapshot of your webpage


Database
information_schema
u301268454_AnthonyMeyer

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B- Create a dBase. Make sure your database's name is your `YourFirstNameLastName`. The username for the database must be `db_YourFirstNameLastName`. For example database: `FaridFarahmand`
User: `db_FaridFarahmand`

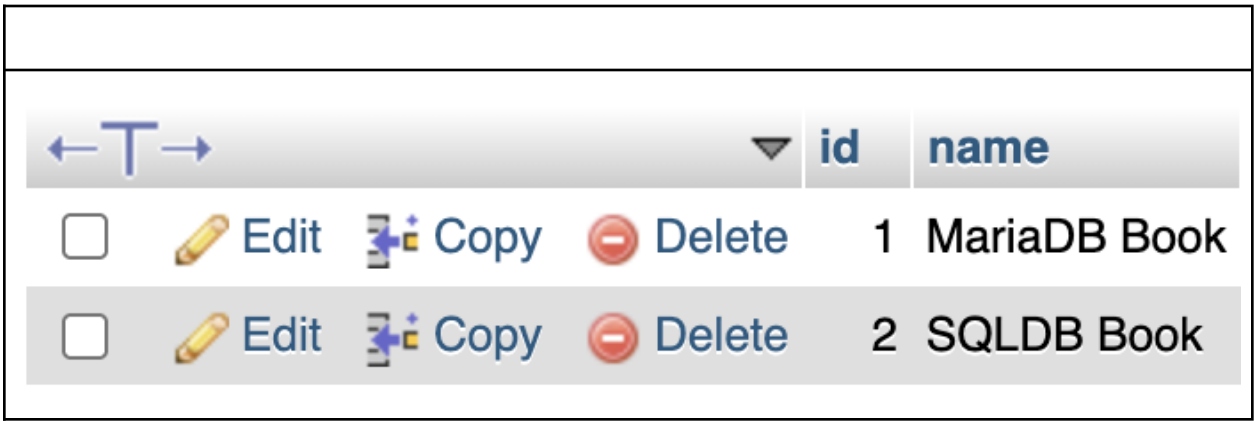
You should see something like this:

Follow the Tutorial and create a simple `BOOK` table. Using `SHOW DATABASES` you can show your dBase. For more information Please refer to Github Tutorial. Your final `BOOK TABLE` should look something like this:

id	name
1	MariaDB Book
2	SQLDB Book

Take a snapshot of your results:

NOTE: Due to limitations on username length (it can only be 14 characters), I had to cut my database username down to **db_LucasLoughn**



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1.2. Practice with MySQL Commands

Using the phpMyAdmin in hostinger.com let's practice more with MySQL commands.

Review the commands in the database tutorial available on the course GitHub. Make sure you complete the tutorial. Once you complete the tutorial you should have a database table with your name and TWO tables having: `Price and Book`. Each table must have the appropriate elements.

Price:

id	price
1	250
2	252
3	223
4	194
5	305

Book:

id	name
1	MariaDB Book1a
2	MariaDB Book2
3	MariaDB Book3
4	MariaDB Book4
5	MariaDB Book5

Now let's combine different fields and create ONE single table showing both book name and price - similar to below:

name	price
MariaDB Book1a	250
MariaDB Book2	252
MariaDB Book3	223
MariaDB Book4	194
MariaDB Book5	305

Take a snapshot of your results: See next page

My Combined Table:

name	price
MariaDB Book1a	250
MariaDB Book2	252
MariaDB Book3	223
MariaDB Book4	194
MariaDB Book5	305

1.3. Create Your Own Sensory-Table

Now you need to design your own database using the following **design requirements**:

Design Two tables: `sensor_data & sensor_resister.`

`sensor_data` must have the following fields:

- node_name <up to 10 characters>; this will be used as the KEY
- time_received <indicating time/date>
- temperature <a numeric value between -10 to +100 celsius with maximum of 6 positions>
- humidity <a number value between 0-100 with maximum of 6 positions>

`sensor_register` must have the following fields:

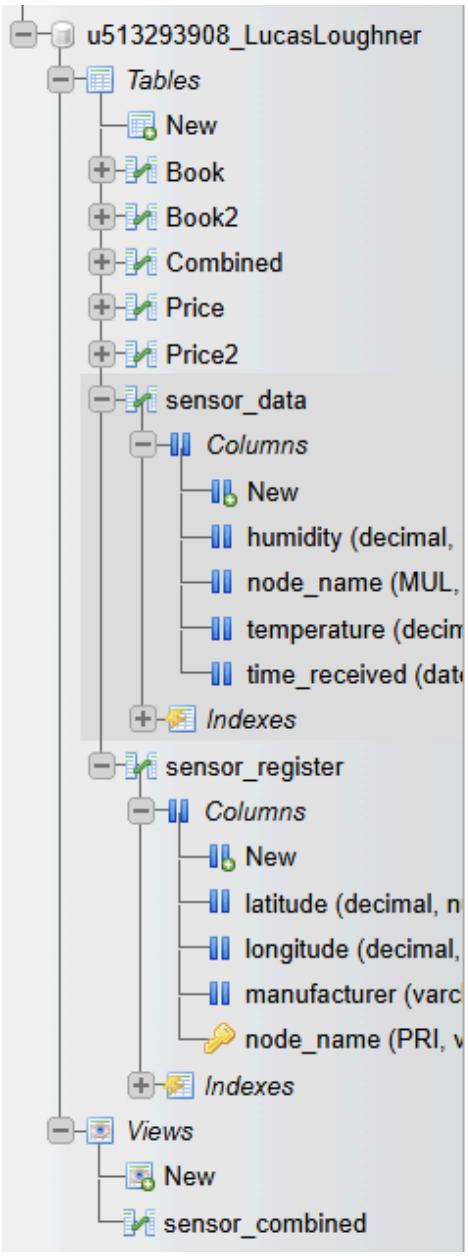
- node_name <up to 10 characters>; this will be used as the KEY
- manufacturer <up to 10 characters>
- longitude < <a real (positive/negative number with maximum of 15 positions>
- Latitude <a real (positive/negative number with maximum of 15 positions>

Your database must have the following requirements:

1- The starting data must be October 1, 2022, at 11:00 am.
2- The data must be sent every half an hour.
3- We need to register 5 different sensors with five different manufacturer names.
5- Each sensor must transmit at least 4 different data sets. This means there must be at least 4 different entries from each sensor node.
6- Node name format must be the following: node_1, node_2, node_3…. Etc.

7- No data from the sensor_data table is accepted if the transmitting node is not registered in the sensor_register table.
8- Create a VIEW of both tables and call it sensor_combined.

Once you have completed the database construction and the VIEW, take a snapshot of the content of the VIEW (sensor_combined) and all the commands you used to construct the databases. In addition, take a snapshot of the database structure, as it appears on PhpMyAdmin portal, as shown below. Your snapshots should be similar to the figures below.



Database as shown on the PhpMyAdmin page

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Creating the Tables:

```sql
-- Table to register sensors
CREATE TABLE sensor_register (
    node_name VARCHAR(10) PRIMARY KEY,
    manufacturer VARCHAR(10),
    longitude DECIMAL(15, 12), -- Max longitude can have 15 digits, with 3 before decimal (180 degrees)
    latitude DECIMAL(15, 12) -- Max latitude can have 15 digits, with 3 before decimal (180 degrees)
);

-- Table for Sensor Data
CREATE TABLE sensor_data (
    node_name VARCHAR(10) NOT NULL,
    time_received DATETIME NOT NULL,
    temperature DECIMAL(6, 3) CHECK (temperature >= -10 AND temperature <= 100),
    humidity DECIMAL(6,3) CHECK (humidity >= 0 AND humidity <= 100),
    FOREIGN KEY (node_name) REFERENCES sensor_register(node_name) -- Only allow sensor data from sensors registered in sensor_register table
);
```

Filling the Tables with Data:

```sql
-- Registering sensors
INSERT INTO sensor_register (node_name, manufacturer, longitude, latitude)
VALUES
('node_1','Siemens', 38.547249, -122.816780),
('node_2','Exxelia', 10.823099, 106.629662),
('node_3','Bosch', 37.774929, -122.419418),
('node_4','Siemens', 38.941631, -119.977219),
('node_5','Exxelia', 11.580230, 108.074181);



-- Creating Sensor Data
INSERT INTO sensor_data (node_name, time_received, temperature, humidity)
VALUES
-- Node 1 (Windsor, CA)
('node_1','2022-10-01 11:00:00',78,45),
('node_1','2022-10-01 11:30:00',83,50),
('node_1','2022-10-01 12:00:00',85,48),
('node_1','2022-10-01 12:30:00',87,55),
('node_1','2022-10-01 01:00:00',92,53),

-- Node 2 (Ho Chi Minh City, Vietnam)
('node_2','2022-10-01 11:00:00',85,85),
('node_2','2022-10-01 11:30:00',88,88),
('node_2','2022-10-01 12:00:00',91,82),
('node_2','2022-10-01 12:30:00',94,87),
('node_2','2022-10-01 01:00:00',95,90),

-- Node 3 (San Francisco, CA)
('node_3','2022-10-01 11:00:00',40,58),
('node_3','2022-10-01 11:30:00',43,66),
('node_3','2022-10-01 12:00:00',46,67),
('node_3','2022-10-01 12:30:00',48,59),
('node_3','2022-10-01 01:00:00',52,62),

-- Node 4 (South Lake Tahoe, CA)
('node_4','2022-10-01 11:00:00',55,33),
('node_4','2022-10-01 11:30:00',56,36),
('node_4','2022-10-01 12:00:00',59,34),
('node_4','2022-10-01 12:30:00',61,38),
('node_4','2022-10-01 01:00:00',62,32),

-- Node 5 (Di Linh, Vietnam)
('node_5','2022-10-01 11:00:00',80,89),
('node_5','2022-10-01 11:30:00',80,88),
('node_5','2022-10-01 12:00:00',78,90),
('node_5','2022-10-01 12:30:00',77,92),
('node_5','2022-10-01 01:00:00',78,87);
```

Complete SQL script to generate the tables^^

DataBase Assignment

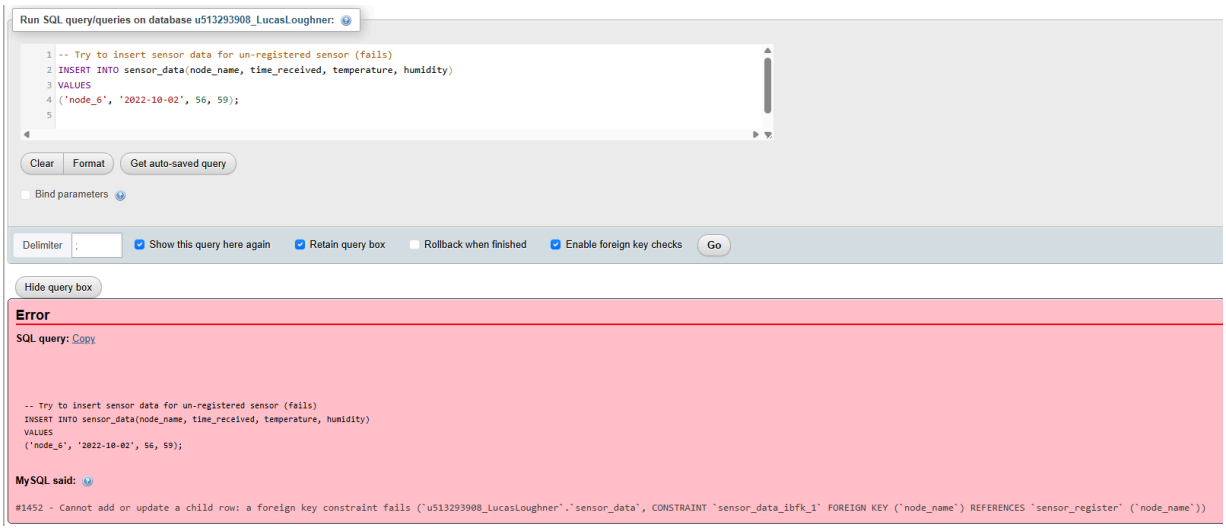
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node_name	manufacturer	longitude	latitude	time_received	temperature	humidity
node_1	Siemens	38.547249000000	-122.816780000000	2022-10-01 11:00:00	78.000	45.000
node_1	Siemens	38.547249000000	-122.816780000000	2022-10-01 11:30:00	83.000	50.000
node_1	Siemens	38.547249000000	-122.816780000000	2022-10-01 12:00:00	85.000	48.000
node_1	Siemens	38.547249000000	-122.816780000000	2022-10-01 12:30:00	87.000	55.000
node_1	Siemens	38.547249000000	-122.816780000000	2022-10-01 01:00:00	92.000	53.000
node_2	Exxelia	10.823099000000	106.629662000000	2022-10-01 11:00:00	85.000	85.000
node_2	Exxelia	10.823099000000	106.629662000000	2022-10-01 11:30:00	88.000	88.000
node_2	Exxelia	10.823099000000	106.629662000000	2022-10-01 12:00:00	91.000	82.000
node_2	Exxelia	10.823099000000	106.629662000000	2022-10-01 12:30:00	94.000	87.000
node_2	Exxelia	10.823099000000	106.629662000000	2022-10-01 01:00:00	95.000	90.000
node_3	Bosch	37.774929000000	-122.419418000000	2022-10-01 11:00:00	40.000	58.000
node_3	Bosch	37.774929000000	-122.419418000000	2022-10-01 11:30:00	43.000	66.000
node_3	Bosch	37.774929000000	-122.419418000000	2022-10-01 12:00:00	46.000	67.000
node_3	Bosch	37.774929000000	-122.419418000000	2022-10-01 12:30:00	48.000	59.000
node_3	Bosch	37.774929000000	-122.419418000000	2022-10-01 01:00:00	52.000	62.000
node_4	Siemens	38.941631000000	-119.977219000000	2022-10-01 11:00:00	55.000	33.000
node_4	Siemens	38.941631000000	-119.977219000000	2022-10-01 11:30:00	56.000	36.000
node_4	Siemens	38.941631000000	-119.977219000000	2022-10-01 12:00:00	59.000	34.000
node_4	Siemens	38.941631000000	-119.977219000000	2022-10-01 12:30:00	61.000	38.000
node_4	Siemens	38.941631000000	-119.977219000000	2022-10-01 01:00:00	62.000	32.000
node_5	Exxelia	11.580230000000	108.074181000000	2022-10-01 11:00:00	80.000	89.000
node_5	Exxelia	11.580230000000	108.074181000000	2022-10-01 11:30:00	80.000	88.000
node_5	Exxelia	11.580230000000	108.074181000000	2022-10-01 12:00:00	78.000	90.000
node_5	Exxelia	11.580230000000	108.074181000000	2022-10-01 12:30:00	77.000	92.000
node_5	Exxelia	11.580230000000	108.074181000000	2022-10-01 01:00:00	78.000	87.000

Snapshot of the VIEW (sensor_combined with all the values in there)

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Prove that if you insert data for a sensor that is NOT registered the value will not be accepted.



Explanation: I try to add sensor data for a sensor named ‘node_6’, which does not exist in sensor_register so it is rejected.

Please note: During your demonstration, you must show that all assigned features are fully functional. Each missing or non-functioning feature will result in a **7-point deduction** from your grade.

Then, in the provided space below, provide the link to that web page. This page will serve as your **main web page for the remainder of the semester**, and all assignment links must be posted there. This is a simple template I have created: https://faridfarahmand.net/SampleStudentAssignmentIndex.html

You can copy the template page and easily update it.

https://lucasloughner.site/
Your web page link.

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2- Answer the following questions

Answer the following questions.

1. What type of database does hostinger.com use?

hostinger uses MariaDB for its databases.

2. What type of web server does hostinger.com use?

hostinger.com supports custom HTML/PHP websites that is highly customizable.

3. Referring to the data type tutorial (https://www.tutorialspoint.com/mariadb/mariadb_data_types.htm) assuming we have the following:

```
CREATE TABLE ints (a INT,b INT UNSIGNED,c INT ZEROFILL);
```

Would this insertion command work? `INSERT INTO ints VALUES (-10,-10,-10);`

Yes, this insertion would technically work, although the actual values inserted into the table would be different from what is requested (b would become 10 instead of -10, for example).

4. What is the purpose of having a primary key in a dBase?

A primary key provides a unique identifier for each row in the database, and helps ensure data integrity. They can also be used alongside foreign keys to establish relationships between different tables in the database.

5. What is the purpose of having a foreign key in a dBase?

The purpose of having a foreign key is to establish a relationship between multiple tables in a database, specifically between the foreign key of one table and the primary key of another table.

6. Can a database have multiple tables?

Yes, a database can (and often does) have multiple tables.

7. What is a relational database?

A relational database is a database comprising multiple, structured tables that are related to each other using primary keys and foreign keys.

8. Is MariadBase a relational or non-relational dBase?

MariaDB is a relational database.

9. What is a database schema?

A database schema is a logical grouping of multiple tables within a database.

10. What is a Table in a relational Database?

A table in a relational database is a structured set of data points with a fixed number of rows and columns. Each row in the table is referred to as a record/tuple, and each column a field/attribute.

11. What are the main components of a Table in a relational database?

The main components of a table are its rows, known as records/tuples, and its columns, known as fields/attributes. A record represents a tuple of values, while each field describes what each value in the record represents. Further, in a relational database, fields can be marked as a "Primary Key" or a "Foreign Key" to establish a relationship with another table.

12. In terms of data type, what is the difference between char(size) and varchar(size)?

char(size) refers to data that must be exactly [size] characters in length, while varchar(size) has the flexibility of being a variable number of characters that is at most [size] characters in length.

13. What would be an example of datetime datatype format?

An example of the format would be '`YYYY-MM-DD HH:MM:SS`', where `YYYY` represents the year, `MM` the month, `DD` the day, `HH` the hour, `MM` the minute, and `SS` the second of the datetime.

3. GitHub

- Please post your code (PHP, SQL, etc.) in your GitHub. Make sure it is clear which assignment the code belongs to. All codes must have comments.
- Please add the link to your GitHub repository on the web page you created in **Section 1.1**.

Link to your GitHub: https://github.com/LukeSev/ECE570-DatabaseAssignmentV2

4. Quick Survey

Answer the following questions, please:

Survey question	Response
On a scale of 1-10 how did you like this exercise? (10 is the best, 1 is the worst)	8
On a scale of 1-10 how much did you learn as a result of completing this exercise? (10 = plenty; 1=very little)	8
How many hours did you spend completing this exercise?	3-4